KAIZEN MATHS

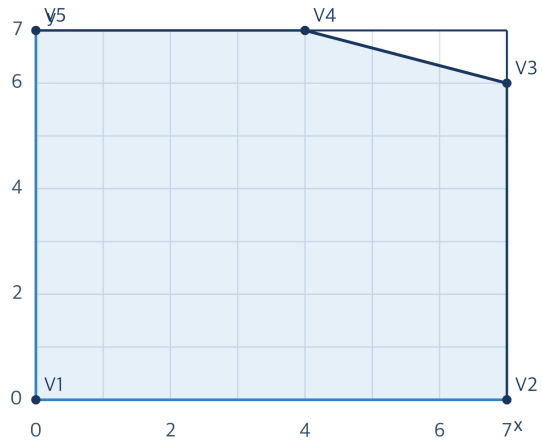
Lesson starter

Linear Programming

Feasible Regions

Name _____

SA1



List the vertices of the feasible region defined by these constraints.

$0 \leq x \leq 7$

$0 \leq y \leq 7$

$x + 3y \leq 25$

ALG-AFEA-L1 · SA

Show your working

KAIZEN MATHS

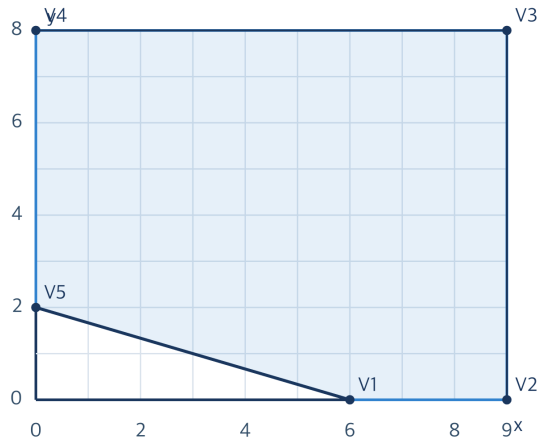
Lesson starter

Linear Programming

Feasible Regions

Name _____

SB1



Decide which of these points are feasible: (6, 0), (4, 4), (10, 5), (0, 0).

$0 \leq x \leq 9$

$0 \leq y \leq 8$

$x + 3y \geq 6$

ALG-AFEA-L1 · SB

Show your working

KAIZEN MATHS

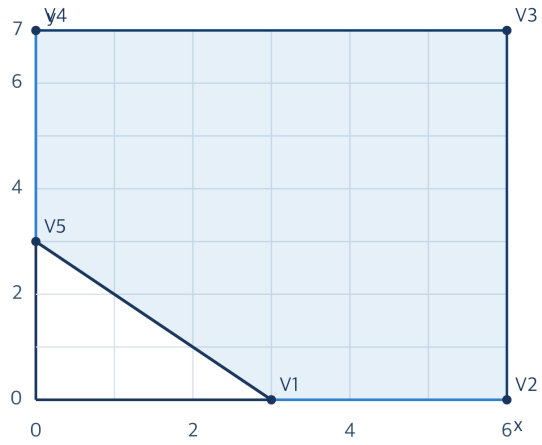
Lesson starter

Linear Programming

Feasible Regions

Name _____

SC1



Describe the feasible region using the constraints and the boundary line.

$0 \leq x \leq 6$

$0 \leq y \leq 7$

$3x + 3y \geq 9$

ALG-AFEA-L1 · SC

Show your working

KAIZEN MATHS

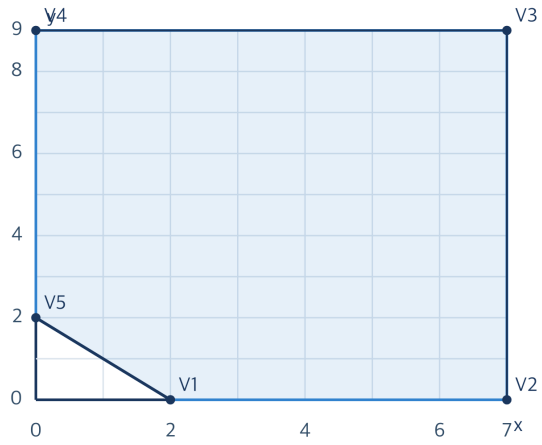
Lesson starter

Linear Programming

Feasible Regions

Name _____

SD1



List the vertices of the feasible region defined by these constraints.

$0 \leq x \leq 7$

$0 \leq y \leq 9$

$2x + 2y \geq 4$

ALG-AFEA-L1 · SD

Show your working

KAIZEN MATHS

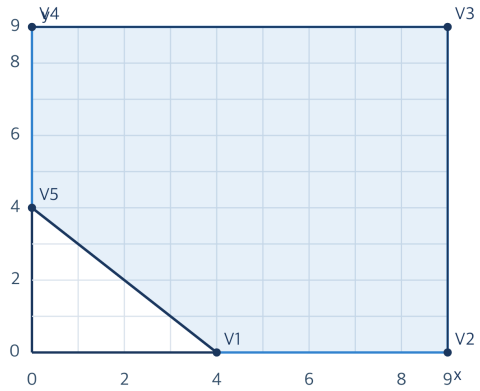
Exit ticket

Linear Programming

Feasible Regions

Name _____

EA1



Decide which of these points are feasible: (4, 0), (4, 4), (10, 5), (0, 0).

$0 \leq x \leq 9$

$0 \leq y \leq 9$

$4x + 4y \geq 16$

ALG-AFEA-L1 · EA

Show your working

KAIZEN MATHS

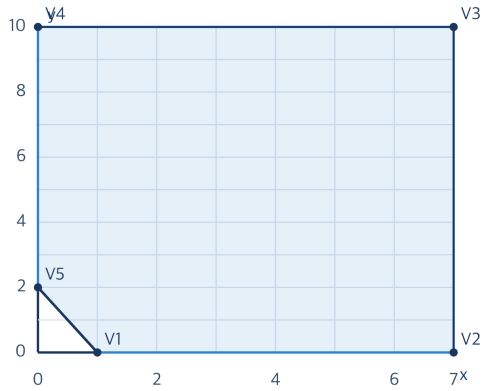
Exit ticket

Linear Programming

Feasible Regions

Name _____

EB1



Describe the feasible region using the constraints and the boundary line.

$0 \leq x \leq 7$

$0 \leq y \leq 10$

$2x + y \geq 2$

ALG-AFEA-L1 · EB

Show your working

KAIZEN MATHS

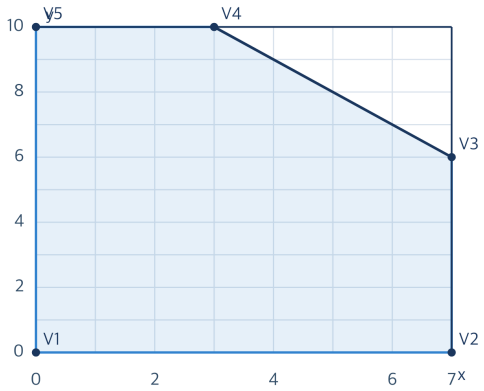
Exit ticket

Linear Programming

Feasible Regions

Name _____

EC1



List the vertices of the feasible region defined by these constraints.

$0 \leq x \leq 7$

$0 \leq y \leq 10$

$4x + 4y \leq 52$

ALG-AFEA-L1 · EC

Show your working

KAIZEN MATHS

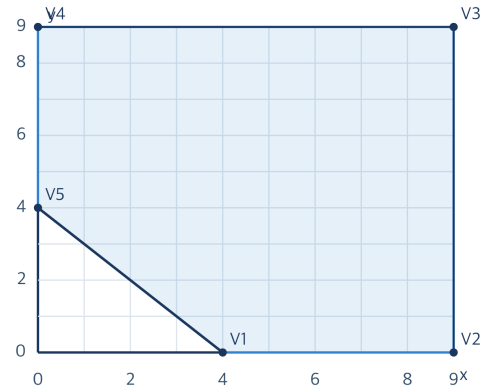
Exit ticket

Linear Programming

Feasible Regions

Name _____

EA1



Decide which of these points are feasible: (4, 0), (4, 4), (10, 5), (0, 0).

$0 \leq x \leq 9$

$0 \leq y \leq 9$

$4x + 4y \geq 16$

ALG-AFEA-L1 · EA

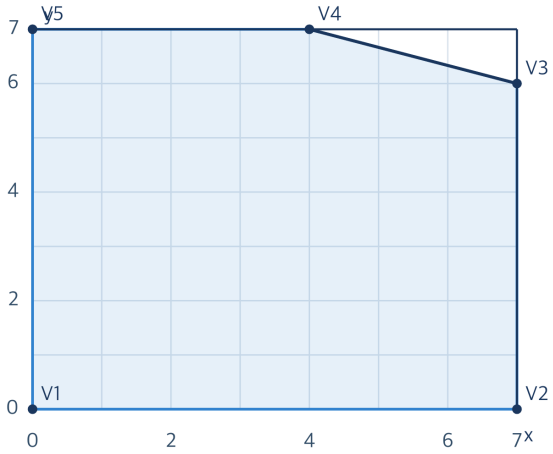
Show your working

Teacher answers & working

Linear Programming · Feasible Regions

ALG-AFEA-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



List the vertices of the feasible region defined by these constraints.

$0 \leq x \leq 7$

$0 \leq y \leq 7$

$x + 3y \leq 25$

Answer:

$(0, 0), (7, 0), (7, 6), (4, 7), (0, 7)$

Working:

The feasible region is the polygon that satisfies all the inequalities at the same time.

The boundary line is x + 3y = 25.

Read each corner of the feasible region from the graph.

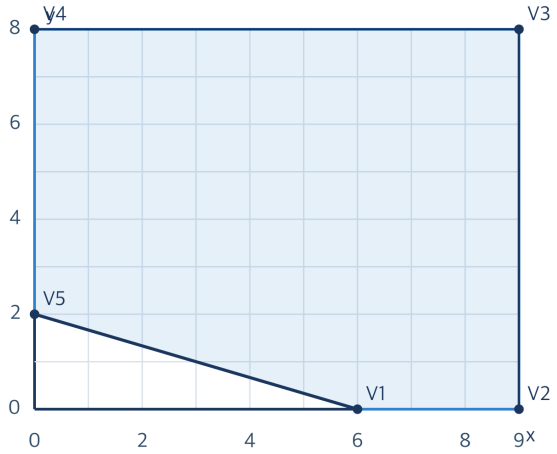
Vertices: (0, 0), (7, 0), (7, 6), (4, 7), (0, 7).

Teacher answers & working

Linear Programming · Feasible Regions

ALG-AFEA-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Decide which of these points are feasible: (6, 0), (4, 4), (10, 5), (0, 0).

$0 \leq x \leq 9$

$0 \leq y \leq 8$

$x + 3y \geq 6$

Answer:

(6, 0), (4, 4)

Working:

A point is feasible only if it satisfies every constraint.

Point	Boundary check	Decision
(6, 0)	$6 \geq 6$	Feasible
(4, 4)	$16 \geq 6$	Feasible
(10, 5)	$25 \geq 6$	Not feasible
(0, 0)	$0 \geq 6$	Not feasible

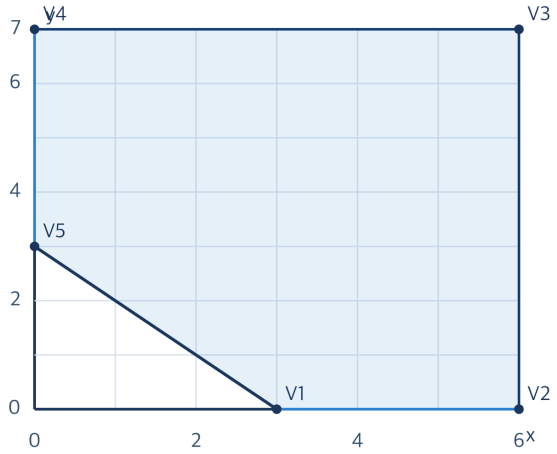
Feasible listed points: (6, 0), (4, 4).

Teacher answers & working

Linear Programming · Feasible Regions

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SC1



Describe the feasible region using the constraints and the boundary line.

$0 \leq x \leq 6$

$0 \leq y \leq 7$

$3x + 3y \geq 9$

Answer:

Inside the box $0 \leq x \leq 6$, $0 \leq y \leq 7$, on the above side of $3x + 3y = 9$.

Working:

The first two constraints keep the region inside the rectangle.

$0 \leq x \leq 6$ and $0 \leq y \leq 7$.

The final inequality keeps the points on the above side of the boundary line.

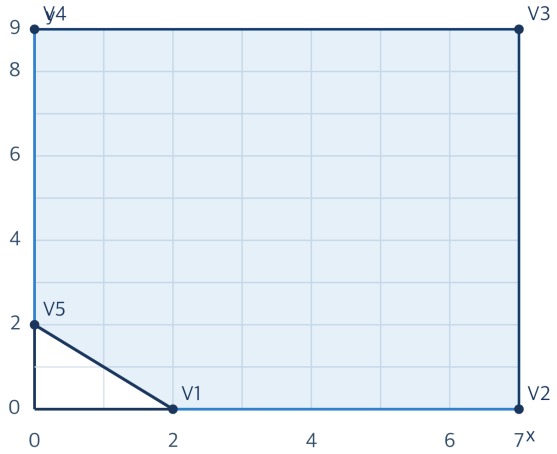
$3x + 3y \geq 9$.

Teacher answers & working

Linear Programming · Feasible Regions

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SD1



List the vertices of the feasible region defined by these constraints.

$0 \leq x \leq 7$

$0 \leq y \leq 9$

$2x + 2y \geq 4$

Answer:

$(2, 0), (7, 0), (7, 9), (0, 9), (0, 2)$

Working:

The feasible region is the polygon that satisfies all the inequalities at the same time.

The boundary line is 2x + 2y = 4.

Read each corner of the feasible region from the graph.

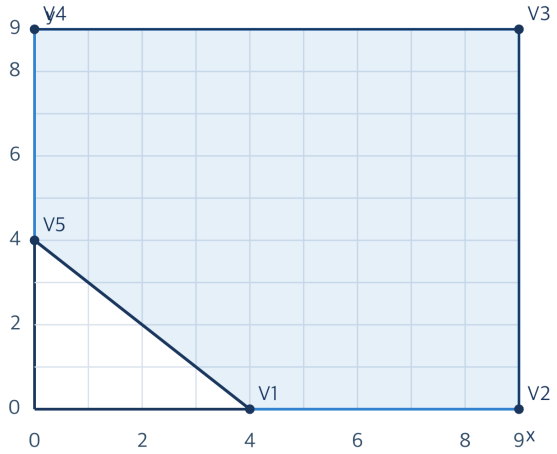
Vertices: (2, 0), (7, 0), (7, 9), (0, 9), (0, 2).

Teacher answers & working

Linear Programming · Feasible Regions

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EA1



Decide which of these points are feasible: (4, 0), (4, 4), (10, 5), (0, 0).

$0 \leq x \leq 9$

$0 \leq y \leq 9$

$4x + 4y \geq 16$

Answer:

(4, 0), (4, 4)

Working:

A point is feasible only if it satisfies every constraint.

Point	Boundary check	Decision
(4, 0)	$16 \geq 16$	Feasible
(4, 4)	$32 \geq 16$	Feasible
(10, 5)	$60 \geq 16$	Not feasible
(0, 0)	$0 \geq 16$	Not feasible

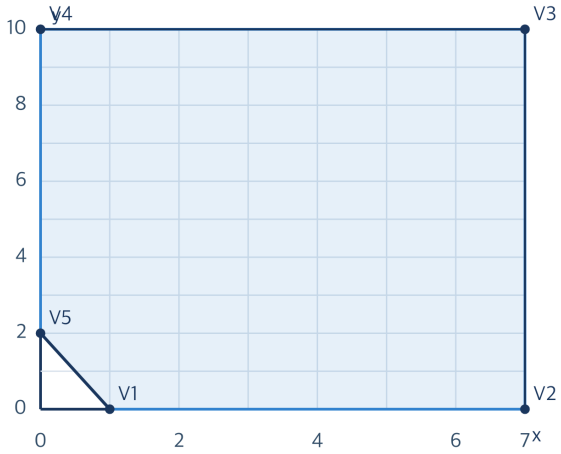
Feasible listed points: (4, 0), (4, 4).

Teacher answers & working

Linear Programming · Feasible Regions

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EB1



Describe the feasible region using the constraints and the boundary line.

$0 \leq x \leq 7$

$0 \leq y \leq 10$

$2x + y \geq 2$

Answer:

Inside the box $0 \leq x \leq 7$, $0 \leq y \leq 10$, on the above side of $2x + y = 2$.

Working:

The first two constraints keep the region inside the rectangle.

$0 \leq x \leq 7$ and $0 \leq y \leq 10$.

The final inequality keeps the points on the above side of the boundary line.

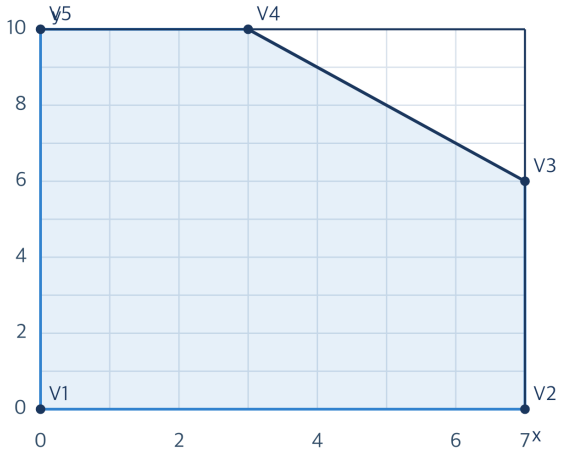
$2x + y \geq 2$.

Teacher answers & working

Linear Programming · Feasible Regions

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EC1



List the vertices of the feasible region defined by these constraints.

$0 \leq x \leq 7$

$0 \leq y \leq 10$

$4x + 4y \leq 52$

Answer:

$(0, 0), (7, 0), (7, 6), (3, 10), (0, 10)$

Working:

The feasible region is the polygon that satisfies all the inequalities at the same time.

The boundary line is 4x + 4y = 52.

Read each corner of the feasible region from the graph.

Vertices: (0, 0), (7, 0), (7, 6), (3, 10), (0, 10).